

Determining if Two Documents are by the Same Author

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Abstract

Almost any conceivable authorship attribution problem is reducible to one fundamental problem: was a pair of (possibly short) documents written by the same author. In this paper, we offer an (almost) unsupervised method for solving this problem with surprisingly high accuracy. The main idea is to use repeated feature sub-sampling methods to determine if one document of the pair allows us to select the other from among a background set of “impostors” in a sufficiently robust manner.

Keywords: Authorship Attribution, Unsupervised Learning

Introduction

The Internet is replete with documents that are written pseudonymously or anonymously and it is often of considerable financial or legal importance to determine if two such documents were in fact written by a single author. For example, we might want to know if several tendentious product reviews were actually by the same, possibly self-interested, writer or, more portentously, if two threatening letters were written by the same assailant. In this paper, we propose a solution to the *authorship verification* problem: determining if two documents were written by the same author. Importantly, we consider cases where the two input documents are not necessarily long.

Note that authorship verification is an open-set problem: we ask if an anonymous document was written by a given candidate author or by anybody else. It is not hard to see that virtually all standard closed-set authorship attribution problems are reducible to the authorship verification problem, while the reverse is not true. In the standard case, we are faced with a closed set of candidate authors for each of whom we have writing samples and are asked to determine which of them is the actual author of some anonymous text. Plainly, if we can determine if any two documents are by the same author, we can solve any such standard authorship attribution problem, regardless of the number of candidates. All we need to do is ask if the anonymous text was written by each of the respective candidates; we will get a positive answer for the true author and a negative answer for all the others. On the other hand, the verification problem is strictly harder than the attribution problem: the fact that we solve a closed-set attribution problem offers no guarantees that we can solve an open-set verification problem. It is thus perhaps not so surprising that, with a single limited exception (see below), no satisfactory solution has previously been offered for the verification problem.

The outline of our solution is as follows. Suppose we are asked to determine if the documents X and Y were written by the same author. We will systematically produce a set of “impostor” documents and – in a matter reminiscent of a police lineup – ask if X is sufficiently more similar to Y than to any of

the generated impostors. The trick is using the proper methods to select the impostors and, more importantly, to measure document similarity. Our measurement of document similarity involves randomly selecting subsets of features that serve as the basis for comparing documents, as will be explained below. We will see that when executed correctly, this method gives surprisingly strong results for the verification problem, even when the documents in question contain no more than 500 words.

In the following section, we briefly review previous related work. In Section 3, we describe the experimental setup and offer two simplistic baseline methods. In Section 4, we define the many-candidates problem and outline its solution. In Section 5, we introduce the impostors method for reducing the authorship verification problem to the many-candidates problem, and in Section 6 we offer results.

Related Work

There has been limited work on the open-set authorship verification problem. Koppel and Schler (2004) introduced the “unmasking” method in which the two input documents are chunked and the effectiveness of machine learning methods at distinguishing them is measured via cross-validation on the chunks. Since chunks of text must be reasonably long (at least a few hundred words) to gain any kind of statistical representativeness, unmasking requires that the input documents be very long. In fact, empirical studies (Sanderson & Guenter, 2006) have shown that unmasking is ineffective for short input documents (less than 10,000 words).

Novak, Raghavan, and Tomkins (2004) considered the case in which the respective writings of 100 authors were each split into two and then needed to be properly matched. They found that using certain feature sets to represent the texts and clustering into 100 pairs yields very strong results. However, in their formulation of the problem, it is known in advance that each document has some match in the set. Furthermore, all 100 matching problems are solved dependently, so that each yields information about the other. Thus, that version of the problem is considerably easier than the one we wish to solve here. Other

work in the same spirit is that of Juola and Baayen (2005) and that of Abbasi and Chen (2008) on what they call the “similarity problem”.

There has also been some work on intrinsic plagiarism detection (Meyer zu Eissen, Stein, & Kulig, 2007) that is similar, though not identical, to the authorship verification problem.

In this work, we will use large sets of impostor candidates. Previous work on large candidate sets for authorship attribution is somewhat limited. Madigan et al. (2005) considered 114 authors, Luyckx and Daelemans (2008) considered 145 authors and Koppel, Schler, and Argamon (2011) considered thousands of authors. Most recently Narayanan et al. (2012) considered as many as 100,000 authors. A few words about authorship attribution with large sets are in order here. The standard authorship attribution in which we need to assign an anonymous document to one of a small closed set of candidates is well understood and has been summarized in several surveys (Juola, 2008; Stamatatos, 2009). As a rule, automated techniques for authorship attribution can be divided into two main types. In *machine-learning* methods, the known writings of each candidate author (considered as a set of distinct training documents) are used to construct a classifier that can then be used to classify anonymous documents (Abbasi & Chen, 2008; Zhao & Zobel, 2005; Zheng, Li, Chen, & Huang, 2006; Koppel, Schler, & Argamon, 2008). In *similarity-based* methods, some metric is used to measure the distance between two documents and an anonymous document is attributed to that author to whose known writing (considered collectively as a single document) it is most similar (Burrows, 2002; Hoover, 2003; [Malyutov 2006](#); [Uzuner & Katz 2006](#); Argamon, 2007; Abbasi & Chen, 2008; [Brennan & Greenstadt 2009](#)). When there are tens, or possibly even hundreds or thousands, of candidate authors, standard machine-learning methods – designed for small numbers of classes – are ~~simply~~ not ~~easily~~ usable. ([In principle, one could use machine-learning methods to learn a separate binary classifier for each candidate author, but this is unwieldy and would in any case require some method for choosing from among multiple positive answers.](#)) In such cases, similarity-based methods are more ~~appropriate-natural~~ than machine-learning methods (Stamatatos, 2009). We will use similarity-based methods in this paper.

Experimental Setup

We will use a corpus consisting of the full output of several thousand bloggers taken from blogger.com. The average blogger in our corpus has written 38 separate blog posts over a period of several years. We will consider pairs of (fragments of) blog posts, $\langle X, Y \rangle$, where X consists of the *first* 500 words produced by a given blogger and Y consists of the last 500 words (on the date we downloaded) produced by a given blogger (who might or might not be the same blogger). We choose the first and last words of bloggers in order to maximize the time gap between the documents we wish to compare; in fact, for the cases in which X and Y are taken from the same blogger, it is never the case that X and Y belong to the same blog post. We choose 500 words per blog to show that our methods are effective even for relatively short document; later we will consider texts of different lengths, both greater and less than 500 words.

We randomly generate a corpus that includes 500 such pairs; for half of them, X and Y are by the same blogger and for the other half, they are not. (No single blogger appears in more than one pair $\langle X, Y \rangle$.) The task is to correctly identify a given pair as *same-author* (i.e., X and Y are by a single blogger) or *different-author* (i.e., X and Y are by two different bloggers).

Note that our problem is ~~essentially an~~ unsupervised ~~problem. We~~ in the sense that we are not supplied with labeled examples of any of the authors in the corpus.

Similarity-Based Baseline Method

Let's first consider two rather simplistic baseline methods for attacking our problem. Given the pair of documents $\langle X, Y \rangle$, the first method is to measure the similarity between X and Y and assign the pair to the class same-author if the similarity exceeds some threshold. This is essentially the method used by Abbasi and Chen (2008) for what they call "similarity detection", though the similarity measures they use are based on features considerably more sophisticated than ours.

To measure the similarity between the documents X and Y , we first represent each document as a numerical vector containing the respective frequencies of each space-free character 4-gram in the document. For our purposes, a space-free character 4-gram is (a) a string of characters of length four that includes no spaces or (b) a string of four or fewer characters surrounded by spaces. We select the 100,000 such features most frequent in the corpus as our feature universe. Character n -grams have long been known to be effective for authorship attribution (Keselj, Peng, Cercone, & Thomas, 2003; Houvardas & Stamatatos, 2006) and have the advantage of being measurable in any language without specialized background knowledge. While other feature sets, such as bag-of-words, function words and parts-of-speech n -grams, are reasonable choices, recent studies (Grieve, 2007; Plackias & Stamatatos, 2008; Luyckx & Daelemans, 2010; Escalante, Solorio, & Montes, 2011) suggest that simpler feature sets such as character n -grams are at least as effective as the alternatives and often even more effective; our preliminary experiments confirmed this finding. In our case, character n -grams have an additional advantage: our method works most naturally with a very large and homogeneous feature set, precisely what is offered by character n -grams. The use of 4-grams, specifically, was found to be effective in experiments reported in Winter (2012).

Let $\vec{X} = \langle x_1, \dots, x_n \rangle$ and $\vec{Y} = \langle y_1, \dots, y_n \rangle$ be the respective vector representations of the documents X and Y , where each x_i represents the *tf*idf* value of a character 4-gram in X and n is the total number of such 4-grams that we consider. We use two standard vector similarity measures, the cosine measure and the min-max measure:

$$\text{sim}(X, Y) = \text{cosine}(\vec{X}, \vec{Y}) = \vec{X} * \vec{Y} / \|\vec{X}\| * \|\vec{Y}\|$$

$$\text{sim}(X, Y) = \text{minmax}(\vec{X}, \vec{Y}) = \frac{\sum_{i=1}^n \min(x_i, y_i)}{\sum_{i=1}^n \max(x_i, y_i)}$$

This baseline method ignores the fact that the similarity of two documents is determined by many factors – genre, topic, etc. – other than author identity; a single uniform threshold for all pairs is not likely to work especially well. In fact, using cosine similarity, the threshold that maximizes accuracy on the development set yields accuracy of 70.6% on the test set. For the min-max similarity measure, the

threshold that maximizes accuracy on the development set yields accuracy of 74.2% on the test set. We will return to these numbers in the results section below.

Supervised Baseline Method

Let's now consider a second baseline method that makes use of a training set. Suppose now that we have a training set consisting of 1000 pairs $\langle X, Y \rangle$, each of which is labeled as a different-author pair or a same-author pair. (The set of authors that appear in this training set are disjoint from those that appear in our corpus.) We will use supervised methods to learn to distinguish between same-author pairs and different-author pairs, as follows. Represent X and Y as vectors, as described above. For a pair $\langle X, Y \rangle$, define $\text{diff}(X, Y) = \langle |x_1 - y_1|, \dots, |x_n - y_n| \rangle$. For each pair $\langle X, Y \rangle$ in the training set, assign the vector $\text{diff}(X, Y)$ the label same-author if $\langle X, Y \rangle$ is a same-author pair and assign the vector $\text{diff}(X, Y)$ the label different-author if $\langle X, Y \rangle$ is a different-author pair. We now use these labeled examples as training examples for supervised learning and apply the learned classifier to our test set. (Note that our classifier learns nothing about specific authors, but rather about what differences in n-gram frequency are characteristic of same-author pairs in general.) Since this is a binary learning problem and SVM has often been found to perform well for binary authorship problems (Abbasi & Chen, 2008; Zheng et al., 2006), we choose SVM as our learning algorithm.

Learning a linear SVM classifier on the development set, exactly as just described, we obtain accuracy of 79.8% on our test set. (This is the strongest result we obtained, using a variety of kernels and parameter setting and various feature sets, including bag-of-words, function words and others; thus, this is the most competitive version of the baseline method against which we compare our algorithm.) We will see below that although our method is almost unsupervised, it performs better even than this supervised method.

The Many-Candidates Problem

We now consider a new approach to the verification problem. Our approach will be based on the solution to a closely related problem: Given a large set of candidate authors, determine which, *if any*, of them is the author of a given anonymous document. We call this problem the *many-candidates problem*; it is sometimes called the *open-set identification problem*. If we can solve the many-candidates problem, we can convert the verification problem into the many-candidates problem by generating a large set of impostor candidates. (Technically speaking, the open-set identification problem is also reducible to the open-set verification problem so these two problems are equivalent. As we saw earlier, closed-set identification is reducible to open-set verification; obviously, it is also reducible to open-set identification.)

Let's consider then how the many-candidates problem can be effectively attacked (Koppel et al., 2011).

In keeping with our experimental setup, suppose that we have a candidate set consisting of 5000 bloggers for each of whom we have the *first* 500 words of their blog. Now we are given the *last* 500 words (which we'll call a *snippet*) of some unspecified blog and are asked to determine which, if any, of the 5000 candidates is the author of this snippet.

Many-Candidates method

Let's start with a somewhat naïve information-retrieval approach to assign an author to a given snippet. Using the feature set and min-max proximity measure defined above, we simply assert that the author of the snippet is the blogger in the candidate set whose text is most similar to the snippet vector. (Note that we use min-max rather than cosine as our proximity measure, because it yielded better results in our similarity-based baseline method above.) Of course, the number of snippets correctly assigned will depend on the length of the snippets and the number of candidate authors. In Figure 1, we show the accuracy obtained for a variety of snippet lengths and sizes of candidate sets. (Each datapoint represents

accuracy obtained for 1000 test snippets.) Thus, for example, we find that when there are 5000 author candidates, each consists of 500 words, 32.5% of the snippets are correctly assigned.

We note that, while 32.5% is perhaps surprisingly high (since the baseline is 0.02%), it is inadequate for most applications. Moreover, this method necessarily assigns every snippet to some author in the candidate set, despite the fact that it might be the case that none of the authors in the candidate set is the actual author. What is required is some criterion by which it can be determined if the best candidate is the actual author of the snippet.

Insert Figure 1 here

As our earlier baseline results indicate, simply requiring that similarity between the best candidate and the snippet exceeds some threshold will not work. Rather, the crucial idea is to vary the feature sets used in representing the texts. If a particular candidate blogger's known text is more similar to the snippet than any other candidate for many different feature set representations of the texts, then that candidate is very likely the author of the snippet. Another candidate's text might happen to be the most similar for one or a few specific feature sets, but it is highly unlikely to be consistently so over many different feature sets.

This observation suggests using the following algorithm (Koppel et al., 2011):

Given: a snippet to be assigned; known-texts for each of C candidates

1. Repeat k times

- a. Randomly choose half of the features in the full feature set.
- b. Find top known-text match to snippet using min-max similarity

2. For each candidate author A ,

- a. $\text{Score}(A) = \text{proportion of times } A \text{ is top match}$

Output: $\arg \max_A \text{Score}(A)$ **if** $\max \text{Score}(A) > \sigma^*$; else Don't Know

The idea is to check if a given author proves to be most similar to the test snippet for many different randomly selected feature sets of fixed size. The number of iterations, k , is a tweakable parameter, but, as we shall see shortly, $k=100$ is sufficient. The threshold σ^* serves as the minimal score an author needs to be deemed the actual author. This parameter can be varied to obtain a tradeoff between recall-precision tradeoff.

This method is similar to classifier ensemble methods in which different classifiers are learned using different subsets of features (Bryll, Gutierrez-Osuna, & Quek, 2003).

Many-Candidates Results

We applied this method to the blogger problem described above, using 1000 test snippets, for various candidate set sizes: 50, 500 and 5000. In Figure 2, we show recall-precision curves generated by varying the score threshold σ^* (where precision is the proportion of correct attributions among all test snippets for which some attribution is given by the algorithm and recall is the proportion of test snippets for which an attribution is given by the algorithm and is correct). As expected, results improve as the number of candidate authors diminishes. We mark on each curve the point $\sigma^*=.80$. For example, for 500 candidates, at $\sigma^*=.80$, we achieve 90.2% precision at 22.2% recall. (Koppel et al. (2011) reported similar results for different candidate set sizes and snippet lengths.)

For the above experiments, we used $k=100$ iterations. We note that using more iterations does not appreciably change results. For example, for the case of 500 candidate authors, recall at 90% precision is 22.3% using 100 iterations; using 1000 iterations it is also 22.3%.

It would be a mistake to conclude, however, that the many-candidates problem is necessarily easier as the number of candidates diminishes. The above result considered cases in which the actual author of a snippet is among the candidate authors. Consider now the possibility that none of the candidate authors is the actual author of the snippet. What we would hope to find is that in such cases the

method does not attribute the snippet to any of the candidates. In fact, testing on 1000 snippets that belong to none of the candidates, we find that at $\sigma^* = .80$, not many are mistakenly attributed to one of the candidate authors: 3.7% for 5000 candidates, 5.5% for 500 and 8.4% for 50. Perhaps counter-intuitively, for snippets by authors not among the candidates, having fewer candidates actually makes the problem *more* difficult since the fewer competing candidates there are, the more likely it is that there is some consistently most similar (but inevitably wrong) candidate. (To take an extreme case, when there are only two candidates, neither of whom is the author, it is plausible that one of them is more similar to the snippet than the other for the preponderance of feature sets; for 1000 candidates, it is unlikely that one of them is consistently more similar than all the others.)

Thus, there is a tradeoff between cases with many candidates (in which case there might be many false negatives) and cases with few candidates (in which case there might be many false positives). It will be important to bear this tradeoff in mind in what follows.

Insert Figure 2 here

The Impostors Method

Let's return now to the verification problem. We are given a pair of documents $\langle X, Y \rangle$ and need to determine if they are by the same author. Since we have seen that we have a reasonably effective solution to the many-candidates problem, we can use impostors to reduce the verification problem to the many-candidates problem. The use of impostors as a background set is a well-established practice in the speaker-identification community (e.g., Reynolds, 1995) and has also been applied to information-retrieval problems (Zelikovitz, Cohen, & Hirsh, 2007), but, as far as we know, has not been previously used for authorship attribution.

We proceed as follows:

1. Generate a set of impostors $Y_1, \dots, Y_m$ (as will be specified below).

2. Compute $score_X(Y)$ = the number of choices of feature sets (out of 100) for which $sim(X, Y) > sim(X, Y_i)$, for all $i=1, \dots, m$.
3. Repeat the above with impostors $X_1, \dots, X_m$ and compute $score_Y(X)$ in an analogous manner.
4. If $average(score_X(Y), score_Y(X))$ is greater than a threshold σ^* , assign $\langle X, Y \rangle$ to *same-author*.

The crucial issues that need to be dealt with are how to choose the impostor set and how many impostors to use. Intuitively, if we choose too few impostors or we choose impostors that are unconvincing – to take an extreme example, imagine X and Y are in English and the impostors are all in Turkish – we will get many false positives. Conversely, if we choose too many impostors or we choose impostors for Y that are in the same genre as X but not in the same genre as Y , we will get many false negatives.

In short, we seek an optimal combination of impostor quality, impostor quantity and score threshold. The three are inter-related. To get some intuition for this, consider three methods of generating a universe of potential impostors for Y :

- **Fixed:** Use a fixed set of impostor documents having no special relation to the document pair in question. For this purpose, we used the aggregate results of random (English) Google queries.
- **On-the-fly:** Choose a variety of small sets of random (medium-frequency) words from Y and use each such set as a Google query; aggregate the top results of the respective queries. This is a set of topically “plausible” impostors for Y . (Of course, the identical procedure is used to generate impostors for X .) The motivation for this method is that it can be applied to any pair of documents on-the-fly with no prior knowledge regarding their provenance. (For our purposes, we use 50 sets of 3 to 5 random words and take the top 25 results for each; since we are satisfied with any plausible impostor set, we make no attempt to optimize these values.)
- **Blogs:** Choose texts from other bloggers. This is a set of impostors that are at least in the same genre as both X and Y . Here we assume that we at least know the shared genre of the pair of documents.

In each of these universes, all documents are of length 500 exactly. To illustrate one key point, we show in Figure 3 the accuracy obtained using the impostors method with varying numbers of impostors drawn respectively from each of the above three universes. Results are shown for the score threshold $\sigma^*=0.10$, but the phenomenon we wish to point out is evident for other threshold values as well.

Insert Figure 3 here

We find that when we choose impostors that are more similar to Y either in terms of genre (Blogs) or content (On-the-fly), fewer impostors are required in order to achieve the same or better accuracy than just choosing random impostors. For all impostor universes, the greater the number of impostors, the more false negatives and the fewer false positives.

In our experiments, we will consider the Blog universe and the On-the-fly universe. In each case, we wish to use reasonably good impostors (to avoid false positives), though not necessarily the very best impostors (to avoid false negatives). We thus use the following protocol for generating impostors (for both On-the-fly and Blogs):

1. Compute the min-max similarity to Y of each document in the universe. Select the m most similar documents as potential impostors.
2. Randomly select n actual impostors from among the potential impostors.

We will see below that results are not particularly sensitive to the choice of m and n .

Results

Blog Corpus Results

For our first experiment, we are given 500 blog pairs as described above and we need to assign each of them to the class *same-author* or to the class *different-author*. We apply five methods, three baseline methods as well as our impostors method with each of two universes. For each method, we use a

parameter to tradeoff recall and precision. Briefly, the five methods and their respective parameters are as follows:

1. Thresholding on similarity using cosine.
2. Thresholding on similarity using min-max.
3. Classifying according to an SVM classifier learned on the training set; signed distance from the boundary is the parameter.
4. The impostors method using the On-the-fly universe; the score threshold is the parameter.
5. The impostors method using Blog universe; the score threshold is the parameter.

Figure 4a shows recall and precision for all methods for the class *same-author* and Figure 4b shows recall-precision curves for the class *different-author*. As can be seen, the impostors method is quite a bit better than the baseline methods, including the supervised method. Also, the Blog universe gives better results than the On-the-fly universe for the impostors method.

Note that for the impostors method using the blog universe, recall at precision=0.9 is 82.5% for the class *same-author* and 66.0% for the class *different-author*. The score threshold σ^* for which we obtain precision of 0.9 for same-author is 0.13; for diff-author we obtain precision=0.9 with score threshold of 0.01. As a practical matter, this means that – assuming a prior probability of 0.5 that X and Y are by the same author – a score above 0.12 indicates that the chance that X and Y are by different authors is less than 10% and a score below 0.02 indicates that the chance that X and Y are by the same author is less than 10%.

Insert Figure 4a here

Insert Figure 4b here

In Figure 5, we show the accuracy obtained at the optimal choice of parameter value for each of our five methods. (For the impostors method, the optimal parameter values are determined on a separate development set that is constructed using the exact same methodology used to construct our test corpus, but on a disjoint set of bloggers. In this sense, our method is technically not completely unsupervised.) We obtain 87.4% accuracy for the impostors method using the Blogs universe. The fact that we obtain 83.2% for the impostors method using the On-the-fly universe is especially encouraging; it means that this method can be successfully applied even in cases where we know nothing at all about the pair of documents, not even their genre.

Insert Figure 5 here

For simplicity, we have shown results for the impostors method using a particular choice of values for the number of potential impostors ($m=250$) and the number of randomly chosen impostors ($n=25$). However, as we can see in Tables 1a (Blog impostors) and 1b (On-the-fly impostors), results for both impostor sets are hardly sensitive to the choices of these parameters as long as each is sufficiently large, although it seems that it is better to randomly choose actual impostors from among the top impostors than to use all the top impostors.

Insert Table 1a Here

Insert Table 1b Here

Note that all our results are for pairs of documents that are of length 500. If we have longer documents, results are even stronger. In fact, as can be seen in Figure 6, accuracy of the impostors method increases as the length of the input documents increases. For documents of length 1500 or greater, accuracy exceeds 90%.

Insert Figure 6 here

All our results thus far relate to a test corpus in which the same-author pairs and different-author-pairs are evenly distributed. In the real world, however, it is typically the case that prior probability that a pair X and Y are by the same author may be very far from 0.5 (in either direction). We therefore show in Table 1 the macro-averaged F1 value obtained for various prior probabilities that X and Y are indeed by the same author. In each case, the score threshold (indicated in the third column of the table) is the one that optimizes macro-averaged F1 (in a separate development corpus) for that particular distribution of same-author and different-author pairs. As the prior probability of *same-author* diminishes, the score we require for a pair to be classed as a same-author pair increases. As can be seen, as long as the prior probability of same-author is not too large, macro-averaged F1 results are quite consistent.

Insert Table 2 here

Student Essays Results

The similarity of two documents by the same blogger across different feature sets presumably results from a common underlying writing style as well as, at least in some cases, the tendency of an individual blogger to often return to particular favorite topics. How much would results be weakened if we systematically guarantee that there is no topical stickiness within individual authors?

To investigate this issue, we considered a corpus of student essays¹ in which each of 950 students has contributed four essays: stream of consciousness, reflections on childhood, an assessment of one's own personality and a thematic apperception test. Our corpus includes 2000 pairs $\langle X, Y \rangle$. (In each case, we use only the first 500 words of the essay.) The critical point is, however, that in every such pair X is chosen from one of these sub-genres and Y is chosen from a different sub-genre. Thus, the topic issue

(and to a lesser extent, the genre issue) is completely neutralized: neither same-author pairs nor different-author pairs are ever about a single topic or in the same sub-genre.

In this case, we choose our impostors for Y from among all those essays that are in the same sub-genre as Y . This method of choosing impostors is akin to choosing blog impostors in the previous problem; we simply leverage what we know about the common nature of a given document pair. In this case, we know only that both are student essays, so we use student essays as impostors. We also know the sub-genre of Y , so we use impostors from its sub-genre.

In Figure 7, we show the accuracy obtained at the optimal choice of parameter value (determined on a separate development set) for each of our four methods on the Student Essay corpus. (There are four methods rather than five in this case because there is only one impostors universe.) We obtain 73.1% accuracy for the impostors method using genre-based-impostors. As can be seen, this result is better than any of the baseline methods, but it is still considerably weaker than the results obtained for the blog corpus. This reflects the fact that same-author pairs in this corpus differ by design both in terms of sub-genre and topic, leaving many fewer common features to exploit. In Figure 8, we show recall-precision curves for each of the four methods. The relative strength of the impostors method is evident.

Insert Figure 7 here

Insert Figure 8 here

Conclusions

In this paper, we~~We have found that even~~considered one of the most fundamental and difficult authorship problems – determining if a pair of short documents is by the same author. ~~—We have found~~that this problem can be solved with reasonable accuracy under certain conditions. This result is of

considerable practical importance, since many real-life problems – for example, authentication of short documents of questionable authenticity – are of this form.

Our approach is almost unsupervised. The method works in two stages. The first stage is to generate a set of impostors that will serve as a background set. We have found that in choosing the impostors one must find the proper balance between the quality of the impostors (that is, their similarity to the “suspect”) and the number of impostors chosen: the more convincing the impostors, the fewer need be used. We have further found that best results are obtained when the impostors are selected from the same genre as the input documents, but that strong results can be obtained even when no information regarding the input documents is available. In such cases, a search engine can be used to generate impostors.

The second stage is to use feature randomization to iteratively measure the similarity between pairs of documents, as proposed in Koppel et al. (2011). If, according to this measure, a suspect is picked out from among the impostor set with sufficient salience, then we claim the suspect as the author of the disputed document. The principle idea is to generate an appropriate set of impostors and then check if elements of the pair pick each other out from among the impostors in the sense of Koppel et al. (2011), namely, by being most similar across a sufficiently large variety of feature set representations. The method works even when we have no prior information about the documents.

There are a number of potential limitations of the method that require further investigation. First, as we have seen, when the two documents in question differ in genre and topic, it is considerably harder to distinguish same-author and different-author pairs.

Another potential pitfall here is that we need to be fairly certain that our impostor documents were not themselves written by the author(s) of the pair of documents in question. This danger does not seem to have adversely affected results in the blog corpus, but it is a potential problem that must be taken into account.

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Selected(n) \ Potential(m)	100	250	500	1000
10	85.6%	87.7%	87.3%	87.2%
25	85.2%	87.4%	87.7%	87.4%
50	85.7%	87.6%	87.7%	87.5%
100	82.8%	86.9%	87.1%	86.8%

Table 1a: Optimal accuracy for various impostor selection configurations using Blog impostors.

Selected(n) \ Potential(m)	100	250	500	1000
10	81.5%	82.5%	82.4%	82.2%
25	83.1%	83.2%	83.2%	83.0%
50	82.1%	83.1%	83.2%	82.4%
100	80.8%	83.1%	82.4%	82.1%

Table ~~24~~b: Optimal accuracy for various impostor selection configurations using On-The-Fly impostors.

Prior	Macro F1	Score Threshold
0.1	86.9	0.30
0.2	86.7	0.22
0.3	87.5	0.19
0.4	87.5	0.18
0.5	87.4	0.14
0.6	86.0	0.14
0.7	83.7	0.05
0.8	81.0	0.05
0.9	75.8	0.01

Table 23: Macro-averaged F1 for various priors for *same-author*.

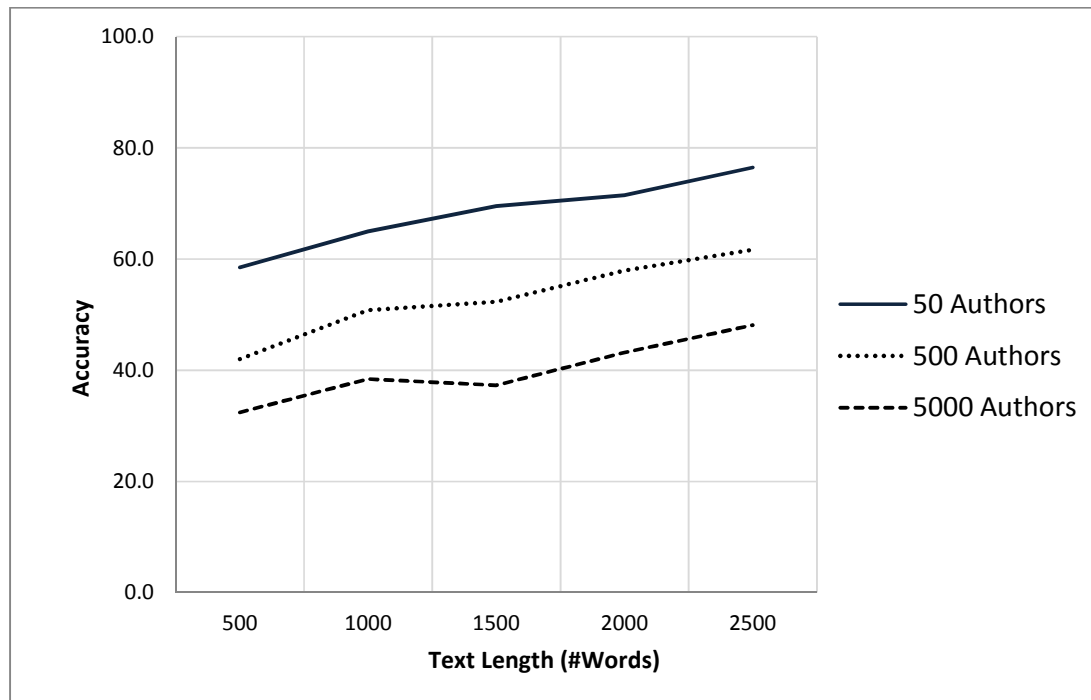


Figure 1: Percentage of correctly assigned snippets per author text length, naïve IR algorithm.

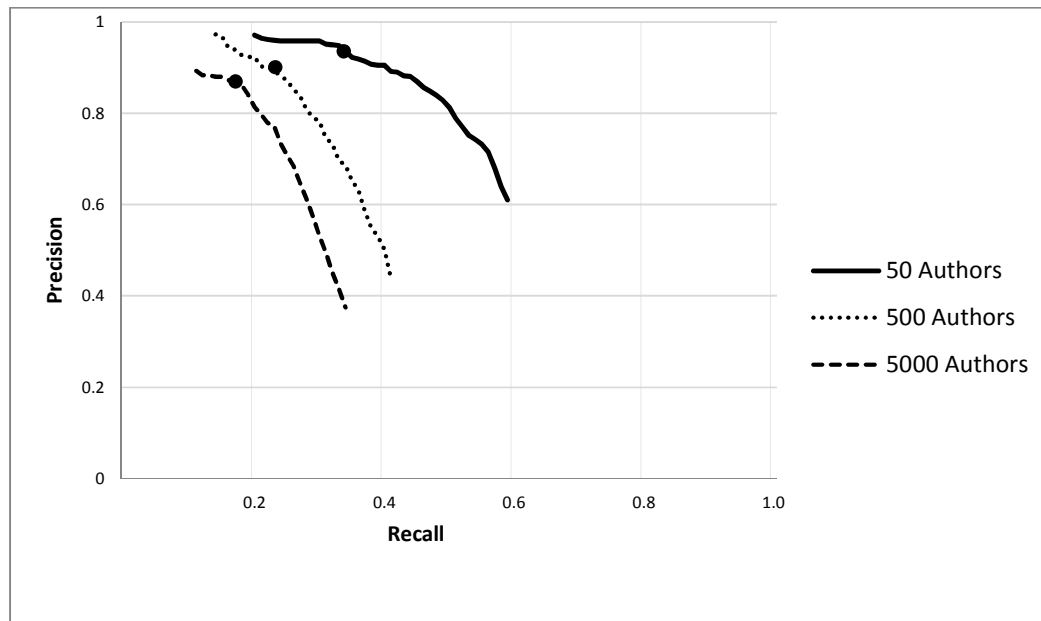


Figure 2: Recall-precision curves for the many-candidates algorithm for various sized candidate sets.

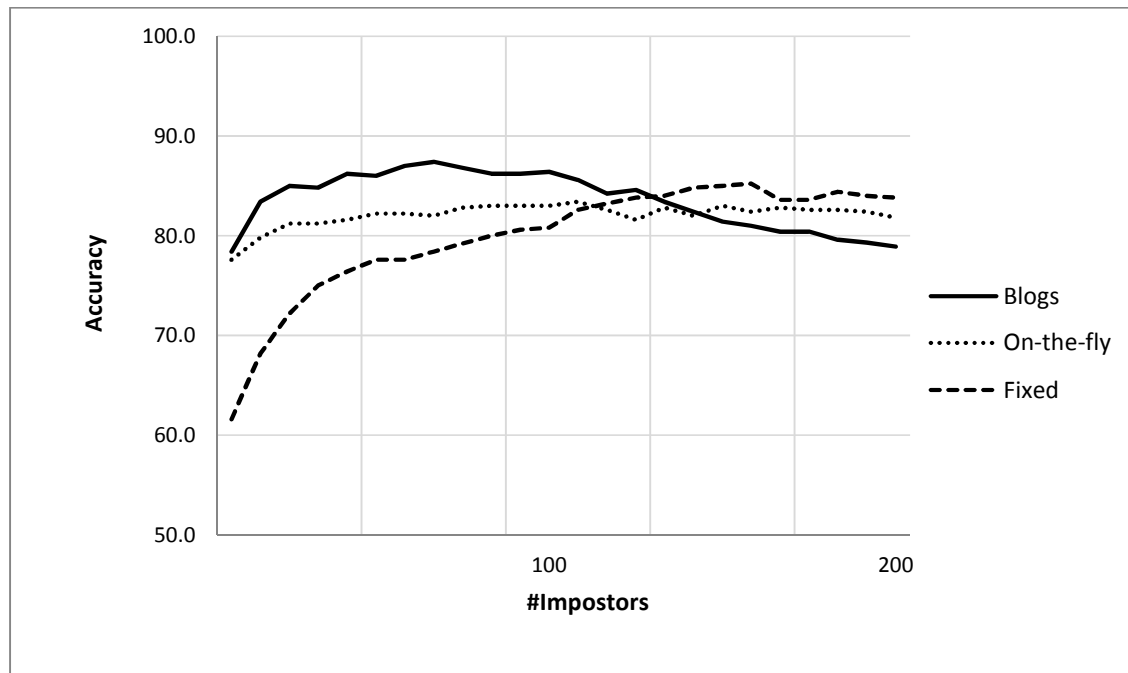


Figure 3: Accuracy of impostors method ($\sigma^*=0.10$) for as impostor set increases for three impostor universes.

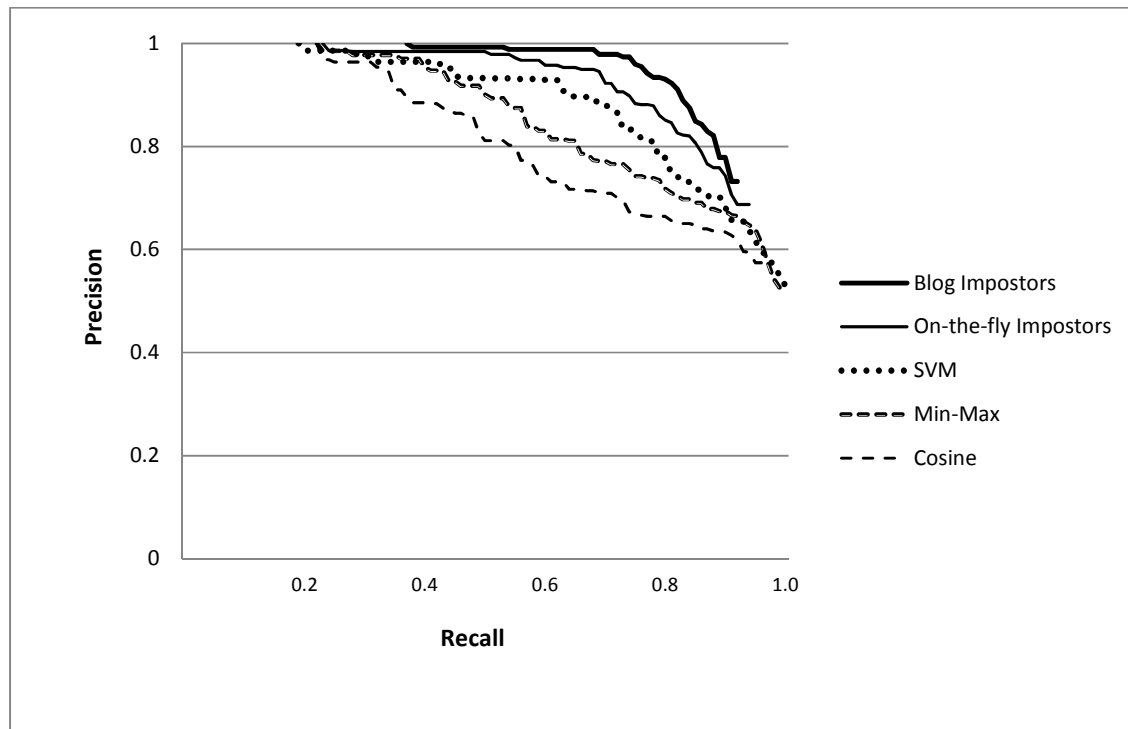


Figure 4a: Recall-precision for the class *same-author*.

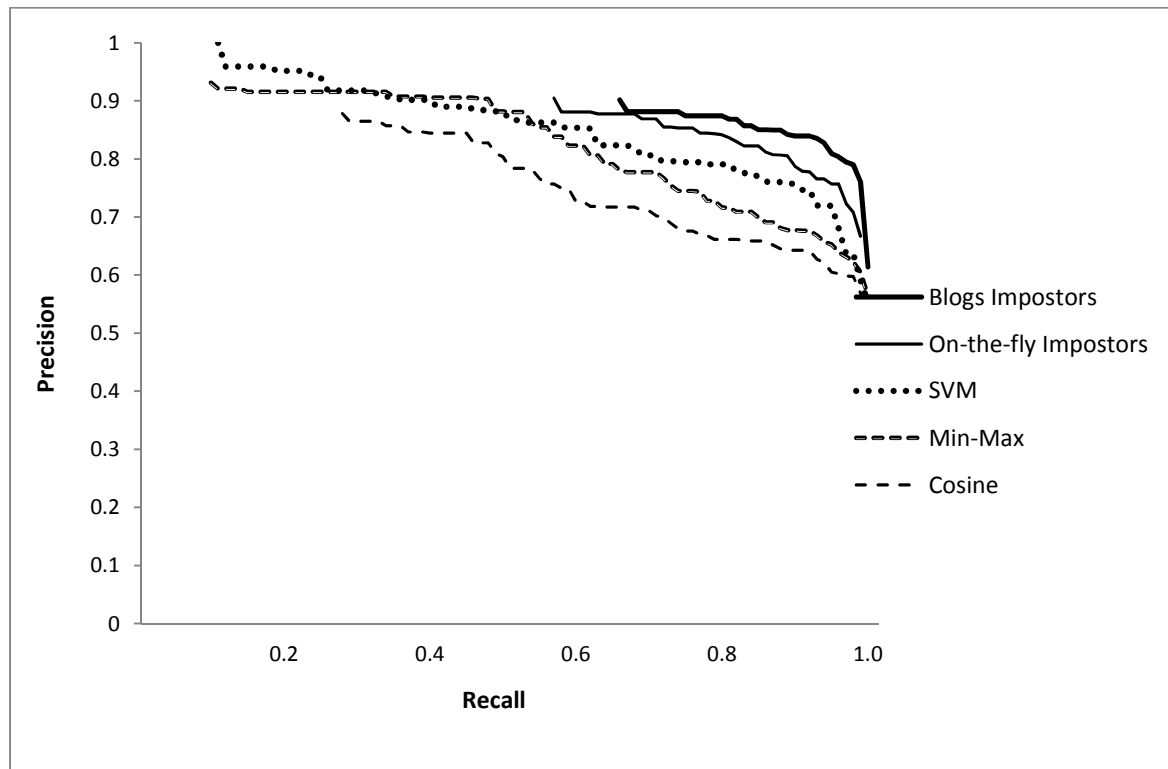


Figure 4b: Recall-precision for the class *different-author*.

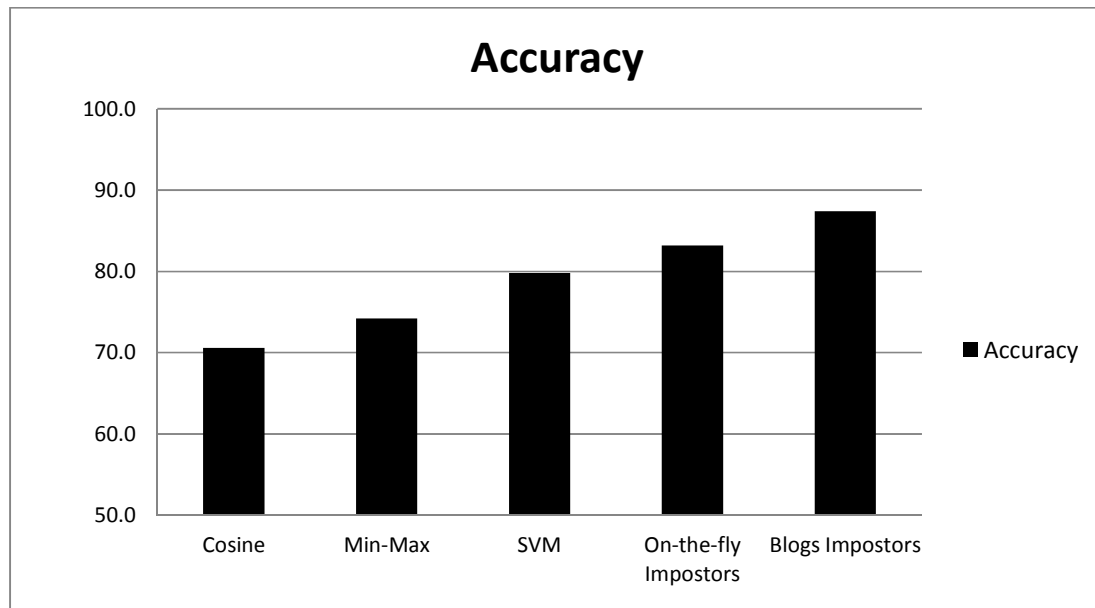


Figure 5: Best accuracy of the five verification methods.

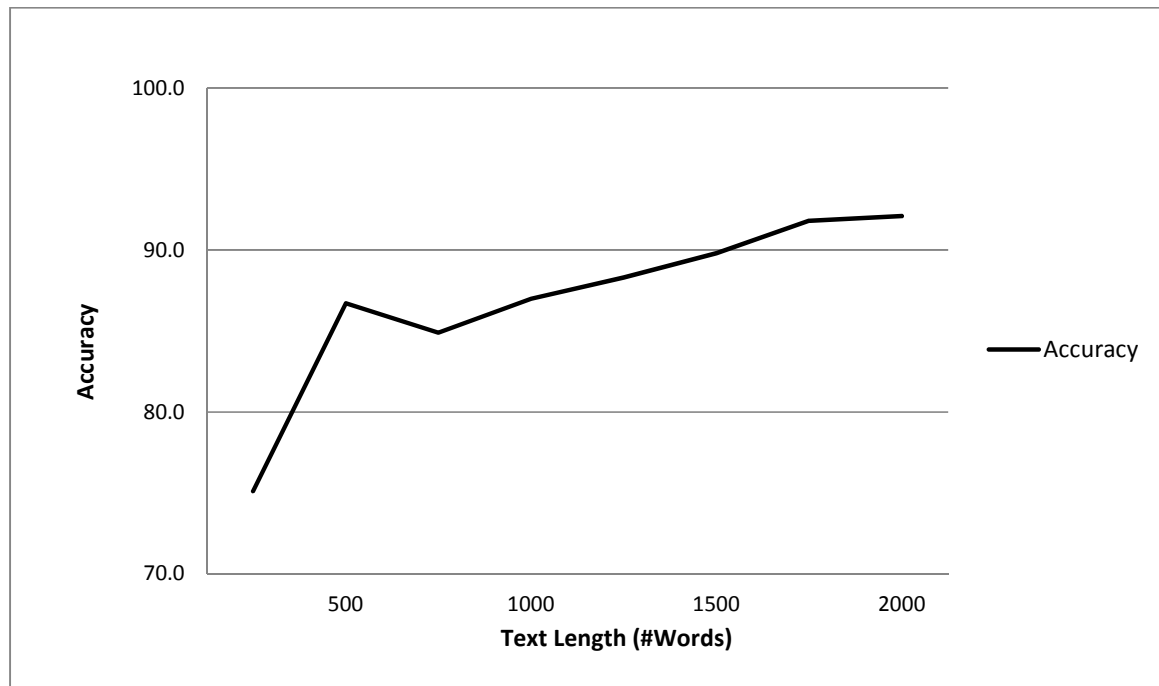


Figure 6: Accuracy as a function of text length, for the impostors-method with blog impostors.

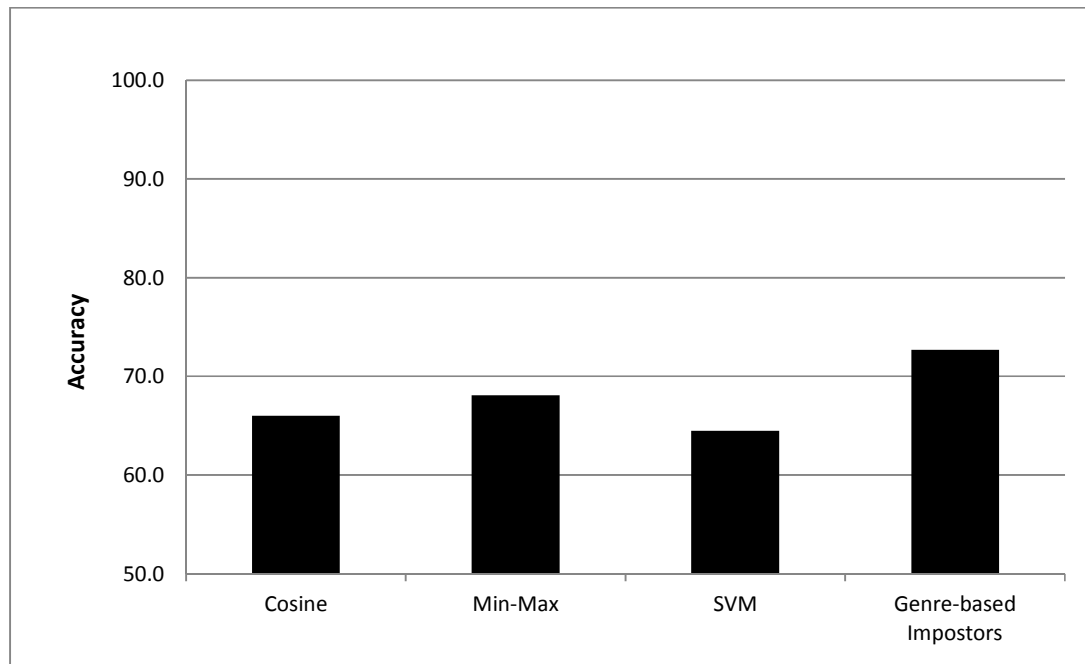


Figure 7: Best accuracy of the various methods on the Student Essays corpus.

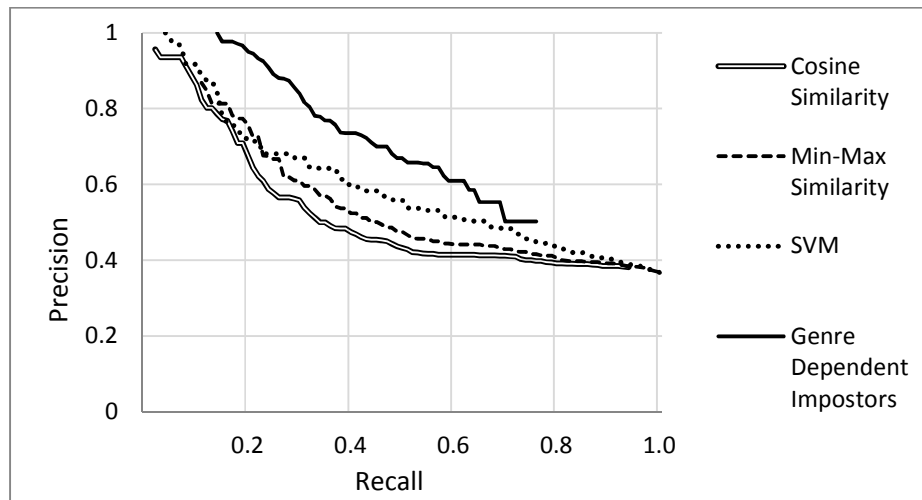


Figure 8: Recall-precision curves for our four verification methods for the Students Essays Corpus for class same-author.

¹ We thank Jamie Pennebaker for making the Students-Essay corpus available to us.